Samuel Yee

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Research Interests

Exoplanet statistics and characterization
Architectures and dynamics of planetary systems

Appointments

University of California, Los Angeles — Los Angeles, CA
51 Pegasi b Postdoctoral Fellow — 10/2025 – present
Supervisor: Erik Petigura

Center for Astrophysics | Harvard & Smithsonian — Cambridge, MA
51 Pegasi b Postdoctoral Fellow — 09/2023 – 09/2025
Supervisor: Dave Charbonneau

Education

Princeton University — Princeton, NJ
Ph.D., Astrophysics; M.Sc., Astrophysics (2020) — 09/2018 – 08/2023
Thesis: *The TESS Grand Unified Hot Jupiter Survey*
Advisor: Joshua Winn

California Institute of Technology — Pasadena, CA
B.S., Physics, with Minor in Geological and Planetary Sciences — 09/2014 - 06/2018

Awards

Heising-Simons 51 Pegasi b Fellowship	2023 – 2027
Princeton University Centennial Fellowship	2018 – 2023
George W. Housner Prize for Academic Excellence and Original Research	2018
Best Poster, Know Thy Star Conference	2017

Publication Summary (see below for full list)

23 major author (1st–3rd author) publications [ADS Library]
43 contributing author publications [ADS Library]

Observing Programs

Summary: I have led as PI or co-PI observing programs across more than 130 nights on various ground-based facilities, including 39 nights on 6–10m class telescopes.

Keck – 6 nights (NASA / NOIRLab)	2021B–2023B
Keck – 6 nights (UC Observatories)	2026A-2026B
Gemini-North – 7.6 nights (NOIRLab)	2024A–2026A
Magellan – 19.5 nights (Princeton / Harvard)	2021B–2025B
WIYN 3.5m – 30 nights (NOIRLab)	2021B–2026B
Lick 2.4m – 6 nights (UC Observatories)	2026B
CTIO 1.5m – 43 nights (NOIRLab)	2021B–2025B
FLWO 1.5m – 12 nights (Harvard)	2024B–2025A
Minerva-Australis – 3 nights (NOIRLab)	2022A–2022B

Advising

Grant Weldon – UCLA PhD Student
Discovery and Characterization of a Puffy Neptune on a Polar Orbit. Paper under review at AJ.

Jea Adams Redai – Harvard PhD Student
Discovery of a Brown Dwarf Transiting a Thick Disk Star. Paper published in ApJ Letters.

Victoria DiTomasso – Harvard PhD Student → Postdoc @ NASA Goddard

A Homogeneous Spectroscopic Analysis of Hot Jupiter Host Stars and New Hot Jupiters around Thick Disk Stars. One paper published in AJ, second paper in prep.

Tarisai Dziire – NYU Shanghai Undergraduate
Characterizing the Hot Jupiter with the Largest TTVs. Paper submitted to AJ.

Grants

Heising-Simons 51 Pegasi b Fellowship	$415k
NASA Keck PI Data Awards (Science PI, 2022B-2023B)	$45k
NASA WIYN PI Data Awards (Science PI, 2021B-2023B)	$39k

Selected Presentations

Exoplanets VI, Porto, Portugal *(contributed talk)*	Jul 2026
57th AAS Division on Dynamical Astronomy Meeting, Chicago *(contributed talk)*	Jun 2026
UT Austin Department of Astronomy Colloquium	Feb 2026
University of Illinois Urbana Champaign Astrophysics Colloquium	Feb 2026
University of Hawai'i Manoa Astronomy Colloquium	Nov 2025
UCLA EPSS Department Colloquium	Nov 2025
UT Austin Department of Astronomy Colloquium	Mar 2025
Know thy Star 2, Pasadena *(contributed talk)*	Feb 2025
Michigan State University Astronomy Colloquium	Nov 2024
Exoplanets V, Leiden *(contributed plenary session talk)*	Jun 2024
Extreme Solar Systems V, Christchurch *(contributed talk)*	Mar 2024
Open Problems in the Astrophysics of Gas Giants, Puerto Natales *(contributed talk)*	Dec 2023
Harvard CfA Exoplanet Pizza Lunch *(invited seminar)*	Oct 2023
AAS General Meeting 241, Seattle *(contributed talk)*	Jan 2023
TESS Science Team Meeting #29, Cambridge, MA *(contributed talk)*	Oct 2022
Harvard CfA Seminar, Cambridge, MA *(invited seminar)*	Oct 2022
MIT TESS Science Talk, Cambridge, MA *(invited seminar)*	Oct 2022
ERES VII, State College *(contributed talk)*	Aug 2022
Exoplanets IV, Las Vegas *(contributed plenary session talk)*	May 2022
Yale Exoplanets and Stars Seminar *(invited seminar)*	Apr 2022
AAS Division on Dynamical Astronomy Meeting 52, online *(contributed talk)*	May 2021

Teaching

Assistant in Instruction	Princeton
AST 205, Planets in the Universe	Sep 2019 – Jan 2020
Teaching Assistant	Caltech
Ph 12c, Statistical Mechanics	Mar 2018 – Jun 2018
Ph 12b, Quantum Mechanics	Jan 2018 – Mar 2018
Peer Tutor	Caltech, 2016 – 2018

Professional Activities

Referee, *AAS Journals*, *MNRAS*, *A&A*, *PASP*	2020 – present
Panel Member, NASA and NOIRLab Time Allocation Committees	2024 – present
External Panelist, JWST and HST Time Allocation Committees	2024 – present
Expert Reviewer, NASA Peer Review Panel	2025, 2026
External Reviewer, Canada Time Allocation Committee	2025
Virtual Organizing Committee, *Emerging Researchers in Exoplanet Science (ERES) V*	2021

Code

Bhatti, W., Bouma, L., & **Yee, S. W.** `cdips-pipeline`: *difference-imaging photometry pipeline*
Yee, S. W., Petigura, E., von Braun, K. `SpecMatch-Emp`: *stellar characterization using an empirical spectral library*

Press Coverage

UCLA: First known 'disappearing planet' has a neighbor that makes it wobble like a top
CNN: This exoplanet is in a death dance with its star
Popular Science: This pitch-black exoplanet is spiraling toward its doom

Outreach & Service

Astronomy.SG: Gave outreach talks organized by Astronomy.SG, a non-profit group focus on astronomy outreach and education in Singapore.

Princeton Public Observing: Organize and host members of the public at observing events using the Princeton department telescope.

Intro2Astro: Gave lectures for Intro2Astro, an online summer course introducing students to astronomy research.

Astro Department Graduate Student Mentorship Program, Princeton University

Graduate Student Buddy Program, Princeton University

Publication List

Updated November 7, 2026

Summary: 66 publications: 23 first–third author; 43 contributing author.
h-index = 18 (10 for leading author publications); 1409 total citations (783 for leading author publications) from ADS.

Major Author Publications [ADS Library]

Students I have advised or co-advised are denoted by †.

23. **Yee, S. W.**, †Dziire, T.., Collins, K., et al. *TOI-3686: A Hot Jupiter and Sub-Neptune Locked in 2:1 Resonance.* submitted to AJ (2026). [arXiv:2610.28533]
22. **Yee, S. W.**, Petigura, E. A., Brahm, R., et al. *A TESS Disappearing Act: The Vanishing Transits of TOI-2535 b.* ApJL, in press (2026).
21. †DiTomasso, V., **Yee, S. W.**, et al. *A Sample of Thick Disk Stars Hosting Hot Jupiters.* AJ, in press (2026). [arXiv:2610.22254]
20. †Monaghan, C., **Yee, S. W.**, et al. *Tidal Rejuvenation: Recent High-Eccentricity Migration as the Origin of WASP-107 b's Inflated Radius Today.* submitted to ApJL (2026).
19. Ma, L., Su, Y., **Yee, S. W.**, et al. *Constraining Tidal Migration with the Hot Jupiter Population.* submitted to MNRAS (2026). [arXiv:2609.10850]
18. †Weldon, G., **Yee, S. W.**, et al. *Discovery of an Inflated Hot Neptune and Its Formation from Jovian Mass Loss.* submitted to AJ (2026). [arXiv:2607.01315]
17. †Adams Redai, J., Chanda, V., **Yee, S. W.**, †DiTomasso, V. et al. *The First Transiting Brown Dwarf Orbiting a Thick Disk Star.* ApJL, 1002, L41 (2026).
16. **Yee, S. W.** & Vissapragada, S. *Popcorn Planets are Not Inflated by Eccentricity Tides.* ApJL, 1000, L54 (2026).
15. **Yee, S. W.**, Tamburo, P., Stefansson, G., et al. *The Polar Orbit of TOI-2374 b, a Planet in the Neptunian Ridge.* AJ, 170, 275.
14. Tamburo, P., **Yee, S. W.**, García-Mejía, J. et al. *Spot-Crossing Variations Confirm a Misaligned Orbit for a Planet Transiting an M Dwarf.* AJ, 170, 200 (2025).
13. **Yee, S. W.**, Winn, J., Hartman, J., et al. *The TESS Grand Unified Hot Jupiter Survey III. Thirty New Planets.* ApJS, 280, 30 (2025).
12. Tamburo, P., **Yee, S. W.**, García-Mejía, J. et al. *The True Stellar Obliquity of a Sub-Saturn Planet from the Tierras Observatory and KPF.* AJ, 170, 34 (2025).
11. **Yee, S. W.**, Stefansson, G., Thorngren, D. P. et al. *The Super-Puff WASP-193 b is on a Well-Aligned Orbit.* AJ, 169, 225 (2025).
10. McKee, B., Montet, B., **Yee, S. W.** et al. *A Planet Candidate Orbiting near the Hot Jupiter TOI-2818 b Inferred through Transit Timing.* ApJ, 981, 106 (2025).
9. **Yee, S. W.**, Petigura, E., Isaacson, H., et al. *Additional Doppler Monitoring Corroborates HAT-P-11c as a Planet.* RNAAS, 8, 187 (2024).
8. **Yee, S. W.** & Winn, J. *The Period Distribution of Hot Jupiters is Not Dependent on Host Star Metallicity.* ApJL, 949, L21 (2023).
7. **Yee, S. W.**, Winn, J., Hartman, J., et al. *The TESS Grand Unified Hot Jupiter Survey II. Twenty New Giant Planets.* ApJS, 265, 1 (2023).
6. **Yee, S. W.**, Winn, J., Hartman, J., et al. *The TESS Grand Unified Hot Jupiter Survey I. Ten TESS Planets.* AJ, 164, 70 (2022).
5. **Yee, S. W.**, Winn, J. & Hartman, J. *How Complete Are Surveys for Nearby Transiting Hot Jupiters?* AJ, 162, 240 (2021).
4. **Yee, S. W.**, Tamayo, D., Hadden, S. & Winn, J. *How Close are Compact Multi-Planet Systems to the Stability Limit?* AJ, 162, 55 (2021).

3. **Yee, S. W.**, Winn, J., Knutson, H., et al. *The Orbit of WASP-12b is Decaying.* ApJL, 888, L5 (2020).
2. **Yee, S. W.**, Petigura, E., Fulton, B., et al. *HAT-P-11: Discovery of a Second Planet and a Clue to Understanding Exoplanet Obliquities.* AJ, 155, 255 (2018).
1. **Yee, S. W.**, Petigura, E. & von Braun, K. *Precision Stellar Characterization of FGKM Stars using an Empirical Spectral Library.* ApJ, 836, 77 (2017).

Contributing Author Publications [ADS Library]

43. Lakhanpal, S., et al. incl. **Yee, S. W.** *The Hot Neptune Initiative (HONEI): IV. TOI-4129 b, an Adolescent Neptune in the Ridge.* submitted to AJ (2026). [arXiv:2611.03810]
42. Armas, K. E., et al. incl. **Yee, S. W.** *TOI-1943 b: A sub-Neptune at the edge of the hot Neptune desert.* submitted to AJ (2026).
41. Daniels, D., et al. incl. **Yee, S. W.** *TOI-2665 b: A Highly Irradiated Giant in the Hot Neptune Desert.* submitted to AJ (2026).
40. Zhang, E., et al. incl. **Yee, S. W.** *Where Did the Angular Momentum Go? WASP-12 b May Have Originated on a Retrograde Orbit.* submitted to AJ (2026).
39. Ruseva, D., et al. incl. **Yee, S. W.** *A well-characterised eccentric short-period Jupiter orbiting TOI-3309.* submitted to AJ (2026).
38. Mancini, L., et al. incl. **Yee, S. W.** *The Hot Neptune Initiative (HONEI). III. An ultra-hot super Neptune orbiting the metal-rich subgiant star TOI-5646: the largest host for a desert dweller.* A&A, 714, A461.
37. Simon, J. D., et al. incl. **Yee, S. W.** *TOI-7169 b: A Hot Jupiter Transiting a Metal-Poor Star.* AJ, 172, 118 (2026).
36. Lillo-Box, J., et al. incl. **Yee, S. W.** *CAFE follow-up of TESS hot Jupiter candidates left behind: I. Five newly confirmed planets and a false positive.* A&A, 713, A272 (2026).
35. Mantovan, G., et al. incl. **Yee, S. W.** *The GAPS programme at TNG. ?. TOI-1533: a compact system hosting a super-Neptune-mass pair with disparate radii.* A&A, 713, A22.
34. Quinn, S., et al. incl. **Yee, S. W.** *TOI-2494 c and TOI-5143 c: A Hot Saturn and a Hot Jupiter with Interior Planetary Companions.* AJ, 171, 359 (2026).
33. Roth, J. T., Hartman, J. D., Bakos, G. A., **Yee, S. W.**, et al. *The T16 Planet Hunt: 10,000 New Planet Candidates from TESS Cycle 1 and the Confirmation of a Hot Jupiter around TIC 183374187.* ApJS, 284, 19 (2026).
32. Bieryla, A., et al. incl. **Yee, S. W.** *TOI-6692b: An eccentric 130 day period giant planet with a single transit from TESS.* AJ, 171, 203 (2026).
31. Wu, Y., et al. incl. **Yee, S. W.** *Detection of Four Cold Jupiters through Combined Analyses of Radial Velocity and Astrometry Data.* AJ, 171, 189 (2026)
30. Gomez Barrientos, J., et al. incl. **Yee, S. W.** *From Earths to Super-Earths: Five New Small Planets Transiting M Dwarf Stars.* AJ, 171, 99 (2026).
29. Manni, F., et al. incl. **Yee, S. W.** *The Hot Neptune Initiative (HONEI) II. TOI-5795 b: A hot super-Neptune orbiting a metal-poor star.* A&A, 701, 230 (2025).
28. Schulte, J., et al. incl. **Yee, S. W.** *Migration and Evolution of giant ExoPlanets (MEEP) II: Super-Jupiters and Lithium-rich Host Stars.* MNRAS, 543, 292 (2025).
27. Essack, Z., et al. incl. **Yee, S. W.** *Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. VI: Confirmation of a Long-Period Giant Planet Discovered with a Single TESS Transit.* AJ, 170, 109 (2025).
26. Naponeillo, L., et al. incl. **Yee, S. W.** *The Hot-Neptune Initiative (HONEI) I. Two hot sub-Neptunes on a close-in, eccentric orbit (TOI-5800 b) and a farther-out, circular orbit (TOI-5817 b).* A&A, 701, 79 (2025).
25. Howard, A. W., et al. incl. **Yee, S. W.** *Planet Masses, Radii, and Orbits from NASA's K2 Mission.* ApJS, 278, 52 (2025).
24. Brinkman, C. L., et al. incl. **Yee, S. W.** *The Compositions of Rocky Planets in Close-in Orbits Tend to be Earth-Like* AJ, 170, 109 (2025).

23. Hon, M., et al. incl. **Yee, S. W.** *A Disintegrating Rocky Planet with Prominent Comet-like Tails Around a Bright Star.* ApJL, 984, L3 (2025).
22. Van Zandt, J., et al. incl. **Yee, S. W.** *The TESS-Keck Survey XXIV: Outer Giants may be More Prevalent in the Presence of Inner Small Planets.* AJ, 169, 235 (2025).
21. Saunders, N., et al. incl. **Yee, S. W.** *TESS Giants Transiting Giants. VII. A Hot Saturn Orbiting an Oscillating Red Giant Star.* AJ, 169, 75 (2025).
20. Ehrhardt, J., et al. incl. **Yee, S. W.** *Confirmation of four Hot Jupiters detected by TESS using follow-up spectroscopy from MaHPS at Wendelstein together with NEID and TRES.* A&A, 692, 220 (2024).
19. Isaacson, H., et al. incl. **Yee, S. W.** *The California Legacy Survey. V. Cataloging Chromospheric Activity Cycles of Planet Search Stars.* ApJS, 274, 35 (2024).
18. Alqasim, A., et al. incl. **Yee, S. W.** *TOI-757 b: an eccentric transiting mini-Neptune on a 17.5-d orbit.* MNRAS, 533, 1 (2024).
17. Radzom, B., et al. incl. **Yee, S. W.** *Evidence for Primordial Alignment: Insights from Stellar Obliquity Measurements for Compact Sub-Saturn Systems.* AJ, 168, 116 (2024).
16. Dai, F., et al. incl **Yee, S. W.** *An Earth-sized Planet on the Verge of Tidal Disruption.* AJ, 168, 101 (2024).
15. Hacker, A., et al. incl. **Yee, S. W.** *TOI-2374 b and TOI-3071 b: two metal-rich sub-Saturns well within the Neptunian desert.* MNRAS, 532, 1612 (2024).
14. Schulte, J., et al. incl. **Yee, S. W.** *Migration and Evolution of giant ExoPlanets (MEEP) I: Nine Newly Confirmed Hot Jupiters from the TESS Mission.* AJ, 168, 101 (2024).
13. Polanski, A., et al. incl. **Yee, S. W.** *The TESS-Keck Survey XX: 15 New TESS Planets and a Uniform RV Analysis of all Survey Targets.* ApJS, 272, 32 (2024).
12. Louden, E., et al. incl **Yee, S. W.** *A Larger Sample Confirms Small Planets Around Hot Stars Are Misaligned.* ApJL, 968, L2 (2024).
11. Battley, M., et al. incl. **Yee, S. W.** *NGTS-30 b/TOI-4862 b: An ∼1 Gyr old 98-day transiting warm Jupiter.* A&A, 686, 230.
10. Hu, Q., et al. incl. **Yee, S. W.** *The PFS view of TOI-677 b: A spin-orbit aligned warm Jupiter in a dynamically hot system.* AJ, 167, 175 (2024).
9. Rubenzahl, R., et al. incl. **Yee, S. W.** *The TESS-Keck Survey. XII. A Dense 1.8 $R_\oplus$ Ultra-Short-Period Planet Possibly Clinging to a High-Mean-Molecular-Weight Atmosphere After the First Gyr.* AJ, 167, 53 (2024).
8. Delamer, M., et al. incl. **Yee, S. W.** *TOI-4201: An Early M-dwarf Hosting a Massive Transiting Jupiter Stretching Theories of Core Accretion.* ApJL, 962, L22 (2024).
7. Schmidt, S., et al. incl. **Yee, S. W.** *Verification of Gaia DR3 Single-lined Spectroscopic Binary Solutions With Three Transiting Low-mass Secondaries.* AJ, 166, 225 (2023).
6. Psaridi, A., et al. incl. **Yee, S. W.** *Three Saturn-mass planets transiting F-type stars revealed with TESS and HARPS.* A&A, 675, A39 (2023).
5. MacDougall, M. G. et al., incl. **Yee, S. W.** *The TESS-Keck Survey. XV. Precise Properties of 108 TESS Planets and Their Host Stars.* AJ, 166, 33 (2023).
4. Zink, J. K. et al., incl. **Yee, S. W.** *Scaling K2. VI. Reduced Small Planet Occurrence in High Galactic Amplitude Stars.* AJ, 165, 262 (2023).
3. Fraizer, R. C. et al., incl. **Yee, S. W.** *NEID Reveals that The Young Warm Neptune TOI 2076b Has a Low Obliquity.* ApJL, 944, L41 (2023).
2. Essack, Z. E. et al., incl. **Yee, S. W.** *TOI-1075 b: A Dense, Massive, Ultra-Short Period Hot Super-Earth Straddling the Radius Gap.* AJ, 165, 47 (2023).
1. Petigura, E. et al., incl **Yee, S. W.** *Planet Candidates from K2 Campaigns 5-8 and Follow-up Optical Spectroscopy.* AJ, 155, 21 (2018).